

# Assignment 2

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## Analysis of sectors –

In the efficient frontier developed in step 1, there are stocks from different sectors including Technology, Industrials, Healthcare, Consumer Cyclical and Utilities. Technology remained the highest contributor with 3 stocks (NVDA, ANET and AVGO) followed closely by Healthcare (LLY, MCK) and Industrials (PWR, CARR) both at two stocks each.

**For Max Sharpe Ratio** – After optimization when we obtain the weights for the portfolio with maximum sharpe ratio we see that the highest contributors to the portfolio are CARR and AVGO with 0.24 and 0.22 weights. One is an industrial company while the other is from the technology sector. PWR, DECK and MCK are also at nearly equal contributions. The portfolio is balanced on the whole with no one stock dominating significantly.

**For Min Variance** – Via optimization we obtain the portfolio for minimum variance in which the largest contributors by weight of stock are CARR (0.39) and AVGO (0.26). They are followed by LLY at 0.12 weight while the other stocks have comparatively very small contribution. This portfolio has a skewed distribution of weights with a major part coming from CARR only.

## Practical Implications of Results –

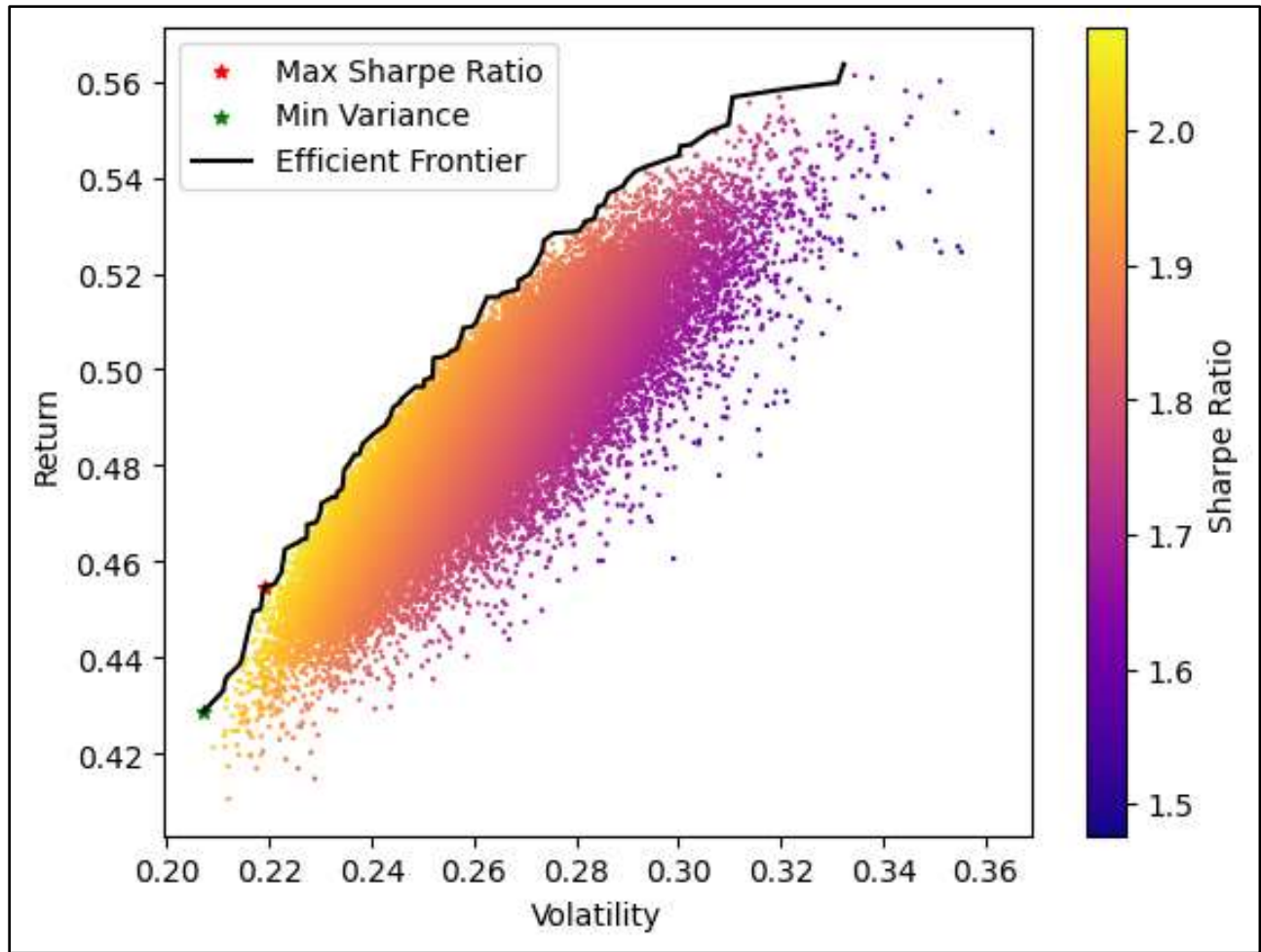
The results obtained from this scipy portfolio optimization analysis have significant practical implications for investment decision-making. By identifying the minimum variance and maximum Sharpe ratio portfolios, we can make informed choices and decisions based on our risk threshold and return target.

The minimum variance portfolio gives the lowest possible risk for a fully invested portfolio, making it the right choice for us if we don't want to take much risk and are looking for a safe bet.

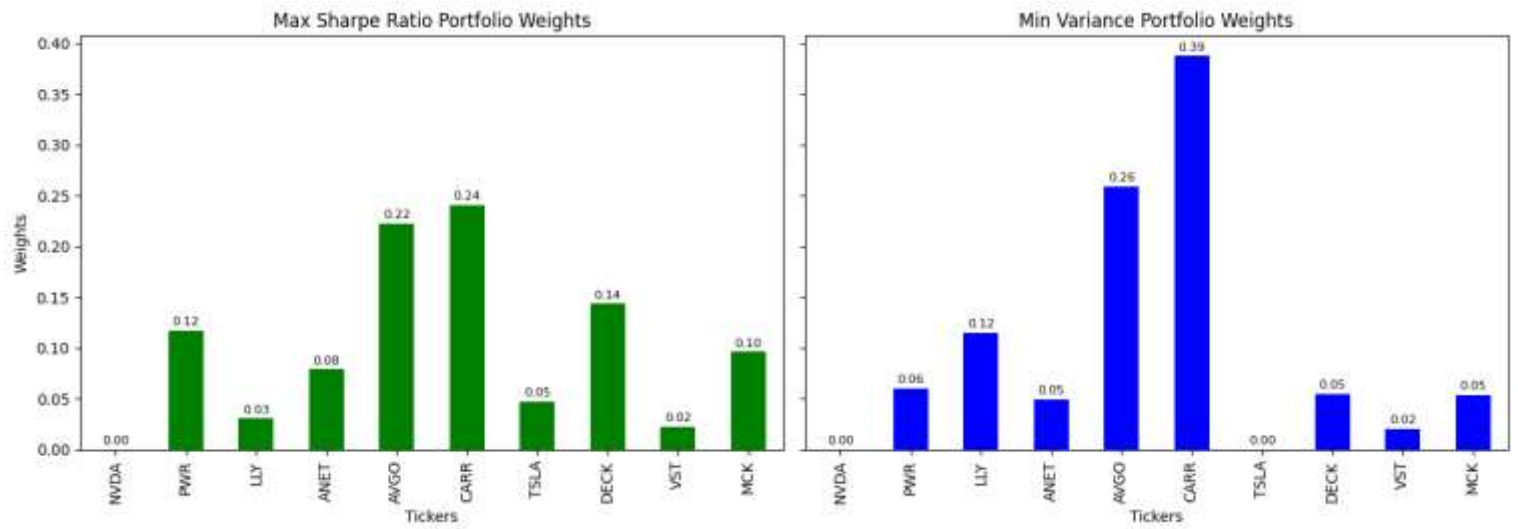
On the other hand, the maximum Sharpe ratio portfolio delivers the highest return per unit of risk, aligning with our goals well if we want to maximize returns.

The graph of the weight distributions further aids in understanding diversification among various sectors and helps reveal which sectors or stocks dominate in these allocations.

## Plots –



Plot of Return versus Volatility indicating the efficient frontier and portfolios with max sharpe ratio and minimum variance.



Plot comparing the optimal weights of different stocks for the portfolio with maximum sharpe ratio and the portfolio with minimum variance.